Research Plan: Nexus of Carbon Reporting and Financial Reporting

Bridging the Synchronization Gap through AI-Driven Forensic Analysis

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1 Context and Research Motivation

The "Synchronization Gap"

There is an important gap between how companies talk about future sustainability and how their past finances are reported. Many APAC firms now announce "Net Zero" goals under IFRS S2 (International Financial Reporting Standard S2 – Climate-related Disclosures) or local rules such as SGX (Singapore Exchange) and HKEX (Hong Kong Exchanges and Clearing), but these promises often do not show up as liabilities under IAS 37 (International Accounting Standard 37 – Provisions, Contingent Liabilities and Contingent Assets). As a result, companies can gain a better reputation without showing the real Capital Expenditures (CapEx) needed in their audited Annual Reports, creating a "Synchronization Gap."

Motivation: Current Research Literature (2024–2025)

This research is motivated by findings from the provided literature, which highlight both US-centric and global challenges:

Information Delays (US Context): Q. Wang (2025) analyzes firms from the NASDAQ-100 and finds that Refinitiv ESG ratings exhibit a significant time-lagged response to ESG-related signals extracted from 10-K filings using ChatGPT-4. The delay is strongest for the Environmental and Social pillars, while Governance shows weaker effects. This pattern is interpreted as being consistent with reputational herding and possible greenwashing, particularly among firms with mid-range ESG ratings.[1]

LLM Analytical Capabilities (Global Context): The FinNLP study (2025) by Xinlin Wang and Mats Brorsson shows that while LLMs such as GPT-4 perform well in narrative financial reasoning, they struggle with precise financial ratio computation in zero-shot settings, particularly when extracting numerical data from heterogeneous formats such as PDFs compared to structured 10-K filings.[5] Related work by Kim et al. (2024/2025) demonstrates that these limitations can be mitigated when numerical

computation is externally assisted via Python scripts and the model is used primarily for analytical reasoning rather than raw arithmetic.[4]

Consequently, we propose building a **Carbon-Financial Synchronization Index (CFSI)** to audit this gap across jurisdictions.

2 Methodology

To address the disparate reporting standards between US and international markets, we adopt a **bifurcated data pipeline**. This design allows us to leverage the structural rigidity of US 10-K filings while accommodating the heterogeneity of APAC and global annual reports.

Phase I: Data Acquisition

Objective: Construct a dataset of **"Carbon-Financial Pairs"** from 100 firms, split between US-dominated majors and international infrastructure leaders.

The US sample will consist majorly of non-financial, carbon-intensive firms in the S&P 500, particularly in sectors such as energy, utilities, materials, industrials, and transportation where sustainability commitments are likely to have material financial implications and where high-quality XBRL(eXtensible Business Reporting Language) data is available.

The international sample will majorly include large-cap, non-financial firms listed on HKEX and SGX operating in emission-intensive sectors such as power generation, cement, steel, chemicals, shipping and heavy manufacturing, where sustainability disclosures are prominent but often weakly reflected in financial provisions.

Pipeline A: US-Dominated Companies (The "Structured" Cohort)

Data Source: 10-K filings filed with the SEC (XML format).

Ingestion Strategy: As demonstrated in Wang (2025), US filings are highly standardized.[1] We utilize Python-based scrapers to parse Item 7 (MD&A) for carbon-related narratives and Item 8 (Financial Statements) for environmental liabilities.

Data collection is conducted programmatically using the SEC's EDGAR (Electronic Data Gathering, Analysis, and Retrieval) system, where financial statements are published in XBRL, a machine-readable tagging format. This approach is analogous to the scripted HTTP-based regulatory data ingestion pipelines I used during my EY internship for large-scale real estate and zoning datasets.

Advantage: The structured nature of 10-K filings enables high-precision extraction of environmental liabilities without reliance on complex OCR pipelines.[1]

Pipeline B: APAC & International Companies (The "Unstructured" Cohort)

Data Source: Annual Integrated Reports and Sustainability Reports (typically PDF) filed under SGX, HKEX, or IFRS mandates.

Ingestion Strategy: As these documents lack the XML structure of US 10-K filings, we employ the PDF-parsing methodology outlined in the FinNLP study.[1] This involves

the use of OCR tools to extract tabular data from the "Notes to Accounts," with particular focus on IAS 37 provisions.[4,5]

Phase II: The Hybrid Analytical Model

Regardless of the data source, all extracted information flows into a unified **Hybrid Analytical Model** to validate the degree of carbon-financial synchronization.

Workflow A: Semantic Verification (The "Narrative" Check)

Semantic Extraction Layer: We isolate a semantic extraction layer that uses ClimateBERT, a climate-domain pretrained transformer trained on a large corpus of climate-related text, to perform sentence-level classification, specificity scoring, and detection of explicit transition-related commitments. This layer outputs structured semantic features and does not perform reasoning, valuation, or financial inference. Following prior evidence that domain-adaptive pretraining substantially improves performance on climate-related classification and specificity tasks, ClimateBERT is used for sentence-level classification, specificity scoring, and detection of explicit transition-related commitments (e.g., presence of net-zero targets, interim milestones, and qualitative pathway descriptions).[6]

Unlike GPT, ClimateBERT is not prompt-based. We pass sentence-level disclosure text through its tokenizer and model to obtain classification and specificity scores. These scores are aggregated deterministically to the firm-year level. Only the resulting structured features are then provided to GPT for limited interpretive mapping, while all financial calculations are performed outside any language model.

Reasoning Layer: Following Kim et al. (2025), we employ GPT-4 with Chain-of-Thought (CoT) prompting.[1] The model is prompted to reason sequentially as follows:

Identify Net-Zero Target Year $\rightarrow$ Identify Engineering Method for Decarbonization $\rightarrow$ Output Implied Liability

Cross-Regional Nuance: For US firms, the model evaluates disclosures using SEC materiality standards. For APAC firms, it assesses constructive obligations under IFRS, i.e., specifically, whether sustainability disclosures are concrete enough to create a reasonable expectation of financial consequences, rather than remaining purely symbolic.

Outputs:

- $N_{\textbf{specificity}}$ **(Semantic Specificity Score):** Measures how concrete a sustainability commitment is.
  - High when disclosures include clear timelines (e.g., "Net Zero by 2035"), explicit engineering methods, and operational detail.
  - Low when language is vague or aspirational.
- $L_{\textbf{shadow}}$ **(Implied / Shadow Liability):** Represents the implied financial obligation if the disclosed transition plan were implemented.
  - A modeled estimate, not an accounting number.
  - Derived from target year, technology pathway, and scale of operations.

Workflow B: Quantitative Verification (Carbon-Adjusted DuPont Analysis)

To avoid the *"math deficit"* of LLMs, Python scripts can execute the ratio analysis. As detailed in the FinNLP study, DuPont Analysis decomposes Return on Equity (ROE) into three drivers. We propose to adjust all three components to reflect the *"Unmanaged Risk"* identified by Wang (2025).

Leveraging my previous experience on DuPont Analysis on the Financial Statements of Zydus Wellness Ltd. as part of a project in a Managerial Accounting course, I propose the following hypothesis: ***(The exact form of these ratio adjustments is tentative and may be revised as the research progresses and better empirical insights emerge)***

1. Net Profit Margin Adjustment (Profitability Impact)

Standard profit margins ignore future carbon costs. We define **adjusted net income** by subtracting the shadow liability (L_{shadow}) that exceeds the reported provision (L_{actual}).

$$\text{Margin}_{\text{reported}} = \frac{\text{Net Income}}{\text{Revenue}}$$

$$\text{Net Income}_{\text{adjusted}} = \text{Net Income} - (L_{\text{shadow}} - L_{\text{actual}})$$

$$\text{Margin}_{\text{adjusted}} = \frac{\text{Net Income}_{\text{adjusted}}}{\text{Revenue}}$$

Rationale: If a firm pledges to decarbonize but books no corresponding cost, its adjusted margin is lower than the reported margin.

2. Asset Turnover Adjustment (Efficiency Impact)

Standard turnover overvalues carbon-intensive assets that should be impaired (*Stranded Assets*). We adjust total assets by subtracting the estimated stranded asset value.

$$\text{Turnover}_{\text{reported}} = \frac{\text{Revenue}}{\text{Total Assets}}$$

$$\text{Turnover}_{\text{adjusted}} = \frac{\text{Revenue}}{\text{Total Assets} - \text{Estimated Stranded Assets}}$$

Rationale: This adjustment shows how efficient a firm would look if assets that may not remain useful in a low-carbon future (such as coal plants) were excluded, rather than treated as an official accounting write-down.

3. Financial Leverage Adjustment (Solvency Impact)

Standard leverage uses reported equity. However, if the shadow liability were booked, retained earnings (and thus equity) would decrease, increasing the firm's true leverage risk.

$$\text{Leverage}_{\text{reported}} = \frac{\text{Total Assets}}{\text{Equity}}$$

$$\text{Leverage}_{\text{adjusted}} = \frac{\text{Total Assets}}{\text{Equity} - (L_{\text{shadow}} - L_{\text{actual}})}$$

Rationale: This reveals the hidden insolvency risk of *"greenwashing"* firms.

Phase III: Index Construction

Objective: Synthesize the findings into the **Carbon-Financial Misalignment Index (CFMI)**.

Distortion Index from DuPont Differences: Let the proportional deviations between reported and adjusted DuPont components be defined as:

$$\Delta\text{Margin} = \frac{|\text{Margin}_{\text{adjusted}} - \text{Margin}_{\text{reported}}|}{|\text{Margin}_{\text{reported}}|},$$

$$\Delta\text{Turnover} = \frac{|\text{Turnover}_{\text{adjusted}} - \text{Turnover}_{\text{reported}}|}{|\text{Turnover}_{\text{reported}}|},$$

$$\Delta\text{Leverage} = \frac{|\text{Leverage}_{\text{adjusted}} - \text{Leverage}_{\text{reported}}|}{|\text{Leverage}_{\text{reported}}|}$$

The **Distortion Index** D is then defined as a weighted aggregation:

$$D = w_m\,\Delta\text{Margin} + w_t\,\Delta\text{Turnover} + w_\ell\,\Delta\text{Leverage}$$

(Weights w_m, w_t, w_ℓ may initially be chosen equal and normalized later.)
The Distortion Index measures the ongoing gap between reported financial results and carbon-adjusted (counterfactual) performance, and serves as a numerical indicator of accumulated sustainability risks that have not been managed.

Index Formulation. An illustrative formulation of the CFMI is:

$$\text{CFMI}_i = \alpha\,N_{\text{specificity}} + \beta\,C_i + \gamma\,T_i + \delta\,D_i$$

Where:

$N_{\text{specificity}}$:= Semantic specificity score derived via LLM analysis

$C_i := \dfrac{L_{\text{actual}}}{L_{\text{shadow}}}$ (financial coverage ratio, lower = larger gap)

T_i := Information lag between narrative signal and rating/accounting change

D_i := Distortion Index from DuPont differences

The weights w_m, w_t, w_ℓ and the parameters α, β, γ are not fixed in advance and will be determined during the course of the research. Their values will be explored using historical data and, where appropriate, simple data-driven or machine-learning–based approaches.

The CFMI is a composite index that quantifies the extent to which a firm's sustainability commitments are not yet reflected in its reported financial performance, indicating accumulated and unrecognized transition risk. A higher CFMI indicates a greater misalignment between a firm's sustainability commitments and its financial recognition.

3 Adaptive Timeline

Phase	Key Activities
Weeks 1–2	**Bifurcated Ingestion**

- US Track: Build 10-K scraper targeting MD&A sections (ref. Wang 2025).[1]
- International Track: Build PDF parser for Annual Reports (ref. FinNLP).[1]
- Checkpoint: Balance data density between XML (US) and OCR (APAC) sources.

Weeks 3–5 **Hybrid Analysis**

- Deploy ClimateBERT and GPT CoT prompts to estimate "Shadow Liabilities" from sustainability narratives.[1]
- Write Python scripts for Carbon-Adjusted DuPont Analysis (margin, turnover, leverage adjustments).[1]
- Checkpoint: Validate shadow liability estimates against a manually curated control set.

Weeks 6–8 **Synthesis & Reporting**

- Estimate the various factors in the formulation of CFMI using Data-driven and ML algorithms and historical data.
- Compute CFMI for various firms.
- Run regression analysis against market volatility to test the "Information Delay" hypothesis.[1]

4 Expected Outcomes

1. **The CFMI Tool:** A versatile Python pipeline capable of auditing both structured US 10-K filings and unstructured international PDF-based reports.
2. **Global Forensic Dataset:** A comparative dataset of "Pledges vs. Provisions" across US and APAC markets.

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